# Year 9 Mathematics – Version 2
# End of Term 3 Test

## Volume, Scientific Notation, Trigonometry and Statistics

Resource Section (Calculators and 1 x A4 (both sides) notes allowed)

Time: 55 mins Name: Solutions / 45 marks

**Full working out must be shown to get full marks. Attempt all the questions.**

### Multi-choice Section **5 marks**

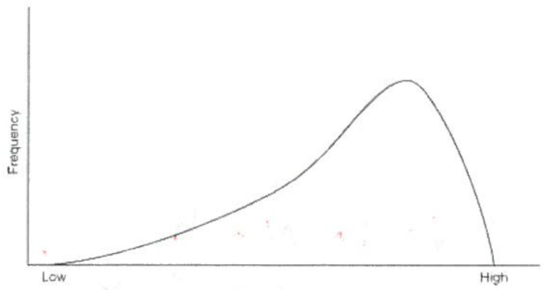
1. **The <u>number of goals</u> scored by each team in a round of netball is an example of:**

   a. Categorical data
   b. Discrete Numerical data
   c. Skewed data
   d. Continuous Numerical data

2. **The MEAN for the set of data [1, 5, 3, 5, 4, 5, 7, 9] is:**

   a. 5
   b. 4.875
   c. 5.571
   d. 8

3. **Describe the shape of the following graph:**




   a. Positively Skewed
   b. Negatively Skewed
   c. Symmetrical
   d. Bimodal

4. **How do we write 6 082 345 in Scientific notation?**

   a. $6.08 \times 10^6$
   b. $6.082345 \times 10^6$
   c. $6.082346 \times 10^6$
   d. $608234.6 \times 10^5$

5. **What would be the volume of a cube, 10 cm x 10 cm x 10 cm?**

   a. 100 $cm^3$
   b. 1000 $cm^3$
   c. 100 $cm^2$
   d. 1000 $cm^2$

# Volume

**6.** Fill in the following table for Units of Volume **5 marks**

$$\begin{tikzcd}
mm^3 \arrow[r, bend left, "\div 1{,}000"] & cm^3 \arrow[l, bend left, "\times 1{,}000"] \arrow[r, bend left, "\div 1{,}000{,}000"] & m^3 \arrow[l, bend left, "\times 1{,}000{,}000"] \arrow[r, bend left, "\div 1{,}000{,}000{,}000"] & km^3 \arrow[l, bend left, "\times 1{,}000{,}000{,}000"]
\end{tikzcd}$$

$$\begin{tikzcd}
mL \arrow[r, bend left, "\div 1000"] & L \arrow[l, bend left, "\times 1000"] \arrow[r, bend left, "\div 1000"] & kL \arrow[l, bend left, "\times 1000"]
\end{tikzcd}$$

**7.** Convert the following: **4 marks**

**a.** 2,400,000 $cm^3$ to 2·4 $m^3$

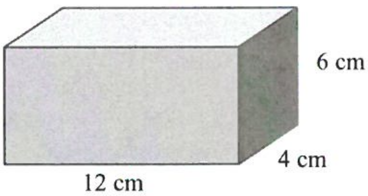
**b.** 2.8 mL to 0·0028 L

**c.** 8.3 $cm^3$ to 8300 $mm^3$

**d.** 0.61 L to 610 mL

**8.** Determine the volume of the following prisms and solid: **6 marks**

**a.**

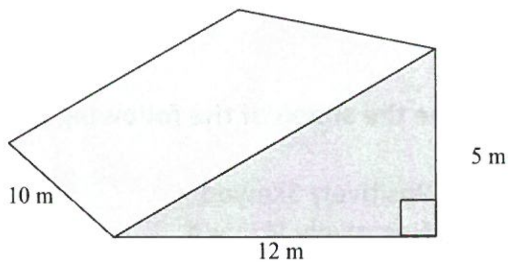



$V = l \times w \times h$

$12 \times 4 \times 6$ ✓

$= 288 cm^3$ ✓

**b.**




**i.** Find the area of the triangle

$A = \frac{1}{2} b \times h$

$= \frac{1}{2} 12 \times 5 = 30 m^2$ ✓

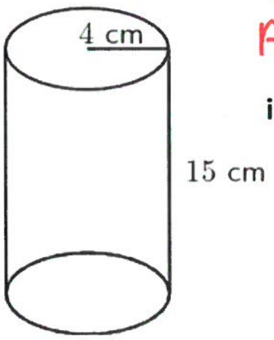
**ii.** Find the volume of the prism

$V = \text{area} \times 10m$

$= 30 \times 10$

$= 300 m^3$ ✓

**c.**




**i.** Find the area of the circle ✓

$A = \pi \times r^2$

$= \pi \times 4^2 = 50{\cdot}27 cm^2$

**ii.** Find the volume of the cylinder

$V = \text{area} \times 15cm$

$= 50{\cdot}27 \times 15$

$= 753{\cdot}98 cm^3$ ✓

## Scientific Notation

**9.** Complete the following table: **4 marks**

| Scientific Notation | Numeral |
|---|---|
| $6 \times 10^1$ | 60 |
| $6 \times 10^2$ | 600 |
| $6 \times 10^3$ | 6000 |
| $6 \times 10^4$ | 60000 |

**10.** Write the following as numerals: **2 marks**

**a.** $3 \times 10^4$ = 30 000

**b.** $7.58 \times 10^{-5}$ = 0.0000758

**11.** Write the following in scientific notation: **2 marks**

**a.** 8, 000, 000 = $8 \times 10^6$

**b.** 0.009802 = $9.802 \times 10^{-3}$

**12.** Solve the following. Write your answer in scientific notation. **2 marks**

**a.** $(2.1 \times 10^2) \times (3 \times 10^3)$

$2.1 \times 3 \times 10^2 \times 10^3$ ✓

$= 6.3 \times 10^5$ ✓

## Statistics

**13.** Frida did a survey on her class and the class next door and recorded the percentage each person received on their latest math test. **4 marks**

**a.** Is this an example of primary or secondary data? Primary ✓

**b.** She presented the data in the graph below. **What specifically is this type of graph called?**

✓ B2B Stem + Leaf

| Class A | | Class B |
|---|---|---|
| 1 5 | 4 | 2 |
| 1 6 8 | 5 | 4 |
| 5 7 7 9 | 6 | 1 6 |
| 6 6 7 8 9 | 7 | 2 5 6 6 |
| 1 2 | 8 | 0 0 4 8 9 |
| 1 | 9 | 3 5 6 7 |

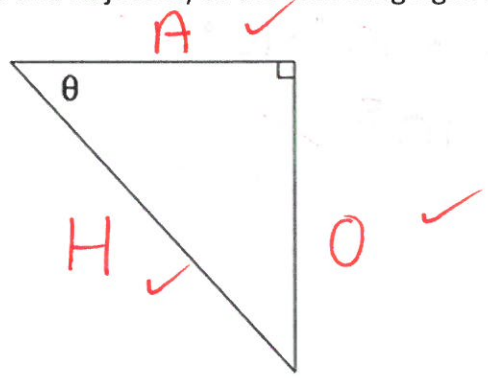
c. Which class do you think did better in the test and explain your answer.

Class B because their results are skewed negatively.

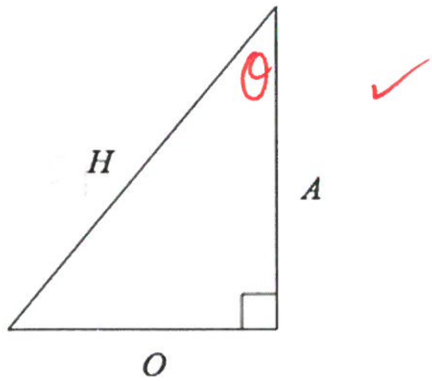
More numbers in th 90s and 80s

**Trigonometry**

**12.** Label the sides (Hypotenuse, Opposite and Adjacent) of the following right-angled triangle. **3 marks**



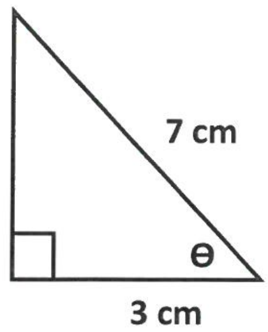

**13.** Label the angle (Θ) in the correct place on the following right-angled triangle. **1 marks**

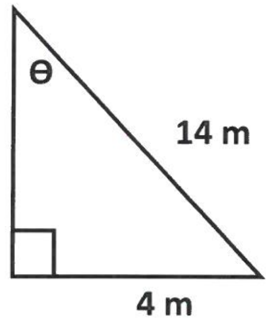


**14.** Match the following Trigonometry Ratios with the correct triangle. **3 marks**

- Hint: Label the sides of the triangles first

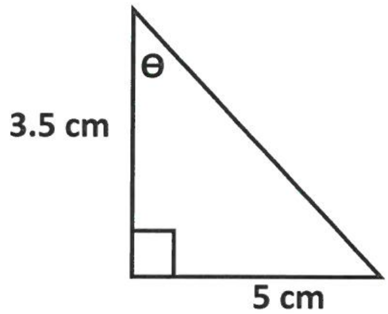
$$\sin\theta = \frac{opposite}{hypotenuse}$$



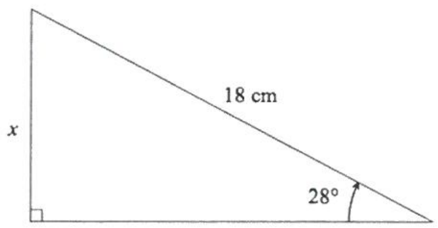

$$\cos\theta = \frac{adjacent}{hypotenuse}$$

$$\tan\theta = \frac{opposite}{adjacent}$$




**15.** Use the **Sin Ratio** to find the length of the missing side in the following right-angled triangle.

**2 marks**




$\sin\theta = \frac{O}{H}$

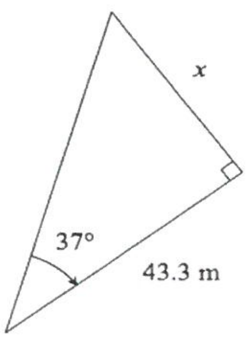
$\sin 28° = \frac{x}{18}$

~~x~~ $\sin 28° \times 18 = x$

$x = 8.45\text{cm}$

**16.** Use the **Tan Ratio** to find the length of the missing side in the following right-angled triangle.

**2 marks**




$\tan\theta = \frac{O}{A}$

$\tan 37 = \frac{x}{43.3}$

$\tan 37 \times 43.3 = x$

$x = 32.63\text{m}$

**End of Test**